NORMALIZATION COMMUTES WITH A SMOOTH MORPHISM UNDER A FINITENESS CONDITION ON COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5409-5415 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5417-5436, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Normalizations, smooth morphisms, and irreducible components of quasi-compact opens; the comparison is retained.

Source reference. Original lemma 0BAL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5436–5436. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 034F: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5436–5436. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 034E: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5436–5436. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 035Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5436–5436. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Normalization and smooth morphisms)**.** *Let $X \to Y$ be a smooth morphism of schemes. Assume every quasi-compact open of Y has finitely many irreducible components. Then the same is true for X and there is a unique isomorphism $X^\nu = X \times_Y Y^\nu$ over X where X^ν, Y^ν are the normalizations of X, Y.*

Proof. By Descent, Lemma 0BAL every quasi-compact open of X has finitely many irreducible components. Note that $X_{red} = X \times_Y Y_{red}$ as a scheme smooth over a reduced scheme is reduced, see Descent, Lemma 034E. Hence we may assume that

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X and Y are reduced (as the normalization of a scheme is equal to the normalization of its reduction by definition). Next, note that $X' = X \times_Y Y^\nu$ is a normal scheme by Descent, Lemma 034F. The morphism $X' \to Y^\nu$ is smooth (hence flat) thus the generic points of irreducible components of X' lie over generic points of irreducible components of Y^ν. Since $Y^\nu \to Y$ is birational we conclude that $X' \to X$ is birational too (because $X' \to Y^\nu$ induces an isomorphism on fibres over generic points of Y). We conclude that there exists a factorization $X^\nu \to X' \to X$, see Morphisms, Lemma 035Q which is an isomorphism as X' is normal and integral over X. $\square$